Run ID: a280_1791577811_9c8912

Adaptive Probe-Guided Genetic Algorithm (APGGS) — Experiment Report

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Engine Version: v4.2

1. Executive Summary & Dataset Metadata

Dataset Name	A280	**Cities / Nodes**	280
Edge Weight Type	EUC_2D	**Known Best Optimum**	2,579
Final Global Best Cost	2,579.00	**Optimality Gap**	+0.0000%
Best Found Generation	15	**Completed Generations**	15
Stagnation Streak	0 / 30	**Stopping Condition**	30 gens without improvement
Experiment Status	COMPLETED	**Total Execution Time**	49.90 s
Stop Reason	0.0% gap achieved (Known optimum found)	**Random Seed**	1

2. Initial High-Quality Route Construction

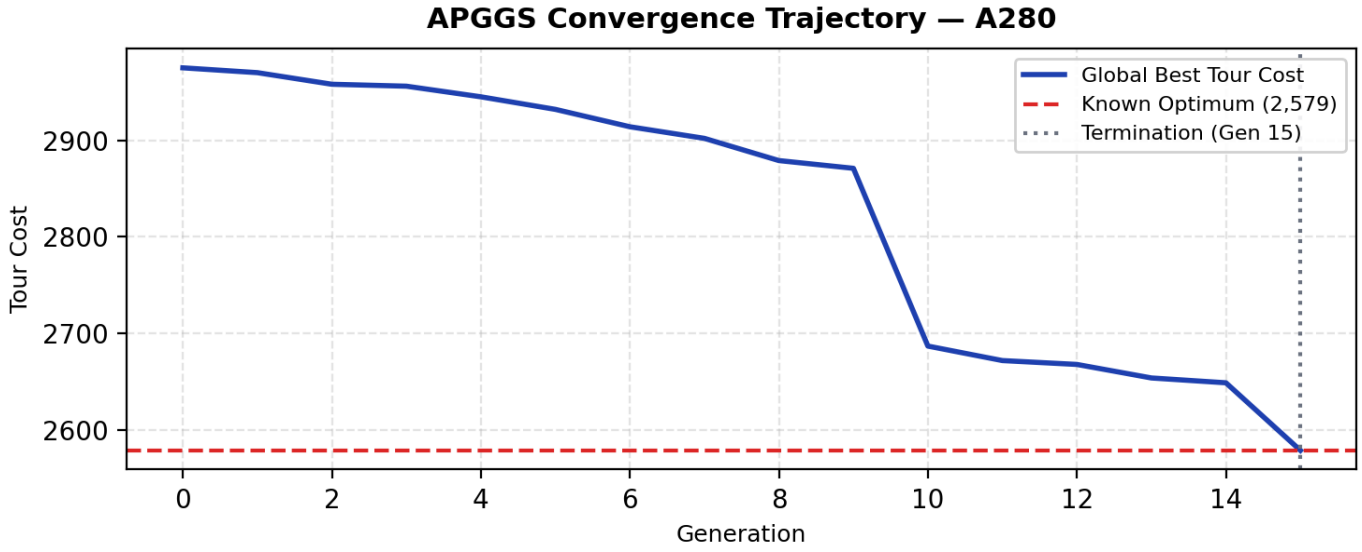
Construction Stage	Resulting Tour Cost	Improvement vs Prev	Duration (s)
Stage 1: Greedy Nearest-Neighbor Tour	3,089.00	Baseline	0.0836
Stage 2: 2-Opt Local Search Improvement	3,086.00	-3.00	0.0002
Stage 3: 2-Opt Deep Refinement Seed	3,050.00	-36.00	0.0007
Stage 4: Initial 70-Individual Population (NN+Greedy+Random)	2,975.00	Population Seeded	3.7469

3. Genetic Algorithm & Adaptive Pipeline Configuration

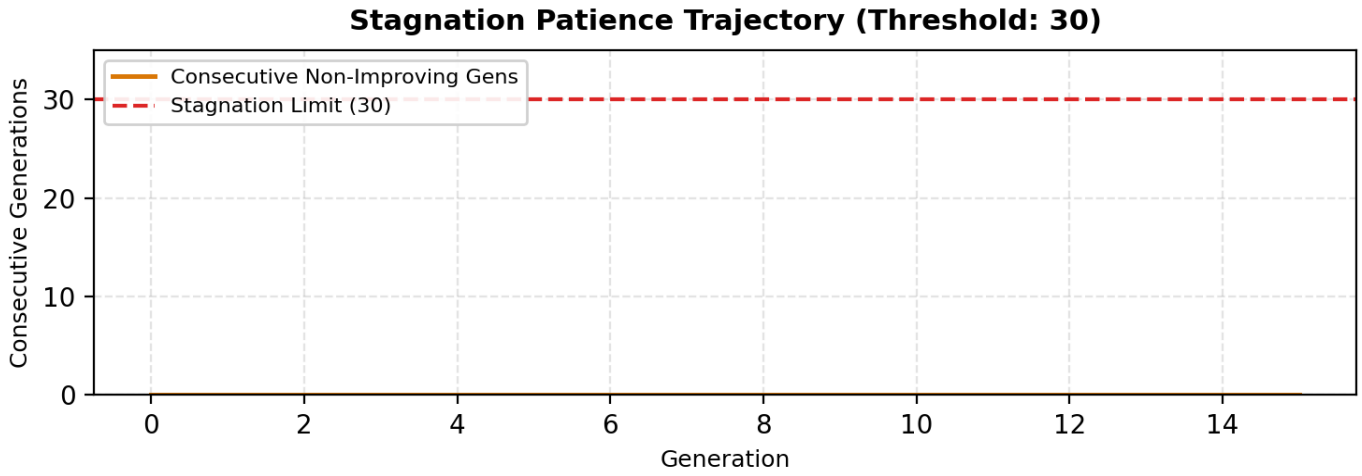
Parameter	Configured Value	Parameter	Configured Value
Population Size	70	Mutation Rate	0.20
Nearest-Neighbor Seeds	17	Swap Fraction	0.02
Greedy Edge Seeds	11	Segment Crossover Prob	0.40
Random Tour Seeds	42	Offspring 2-Opt Prob	0.35
K-Nearest Neighbors	15	Adaptive Elites Count	14
Tournament Size	4	PMG Memory Decay	0.92
Selective LKH Cadence	Every 15 gens	Stagnation Window	15
Selective LKH Individuals	3	Stagnation Patience Limit	30 consecutive gens

4. Research Visualizations & Trajectory Curves

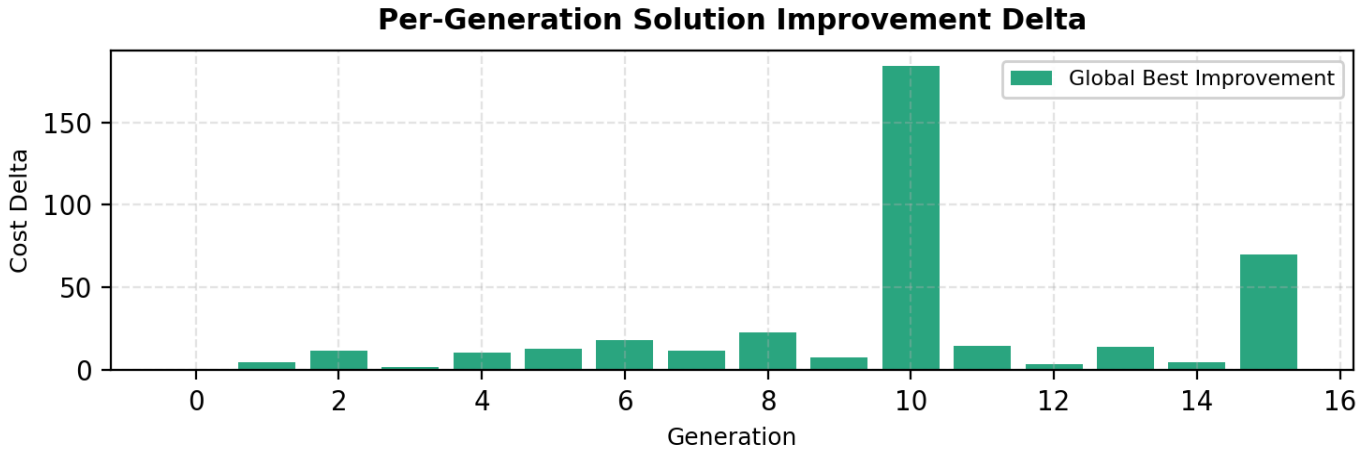
Convergence Trajectory: Global Best Cost vs Generation

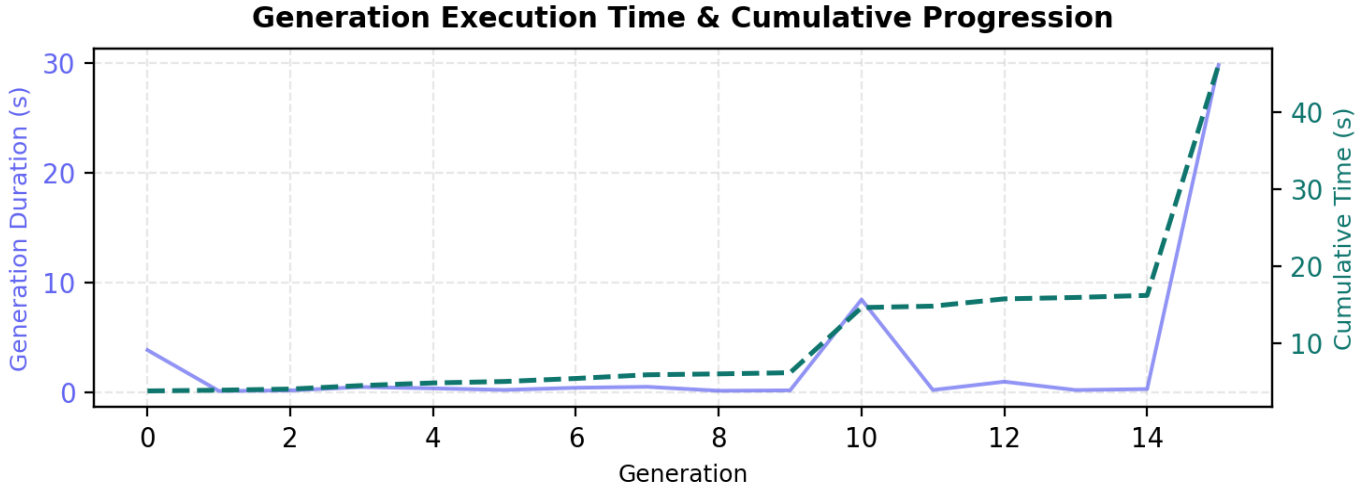


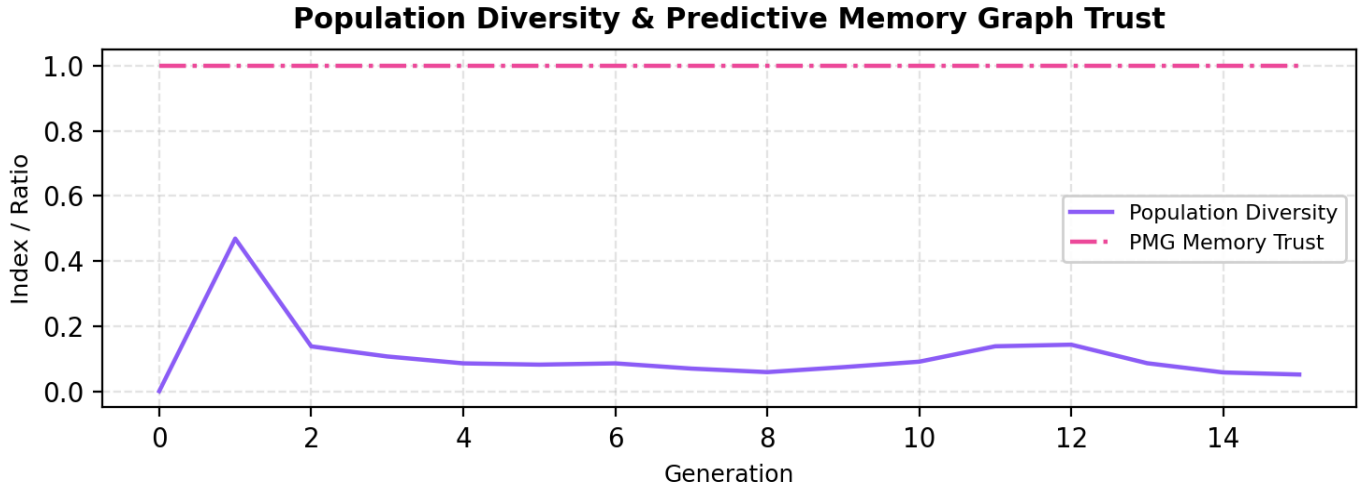
Stagnation Counter Progression (Target: 30 Consecutive Non-Improving Gens)



Per-Generation Cost Improvement & Runtime Progression







5. Evolution Highlights (Global-Best Discoveries)

Gen	Global Best Cost	Cost Reduction	Gap (%)	Operator / Method	PMG Trust	Non-Imp Streak	Duration (s)
0	2,975.00	0.00	+15.3548%	Nearest Neighbor (start=178)	1.00	0	3.8314
1	2,970.00	5.00	+15.1609%	Elite + two_opt	1.00	0	0.0986
2	2,958.00	12.00	+14.6956%	Segment Crossover + Fast 2-Opt + two_opt	1.00	0	0.1455
3	2,956.00	2.00	+14.6181%	Elite + weak_edge_surgery	1.00	0	0.4577
4	2,945.00	11.00	+14.1915%	Elite + weak_edge_surgery	1.00	0	0.3373
5	2,932.00	13.00	+13.6875%	Segment Crossover + Fast 2-Opt + weak_edge_surgery	1.00	0	0.1899
6	2,914.00	18.00	+12.9895%	Segment Crossover + weak_edge_surgery	1.00	0	0.3900
7	2,902.00	12.00	+12.5242%	OX Crossover + Fast 2-Opt + weak_edge_surgery	1.00	0	0.4761
8	2,879.00	23.00	+11.6324%	Segment Crossover + Fast 2-Opt + two_opt	1.00	0	0.1168
9	2,871.00	8.00	+11.3222%	Elite + two_opt	1.00	0	0.1517
10	2,687.00	184.00	+4.1877%	OX Crossover + Fast 2-Opt + predictive_or_opt	1.00	0	8.4451
11	2,672.00	15.00	+3.6060%	OX Crossover + Fast 2-Opt + two_opt	1.00	0	0.1864
12	2,668.00	4.00	+3.4509%	Segment Crossover + Fast 2-Opt + predictive_or_opt	1.00	0	0.9431
13	2,654.00	14.00	+2.9081%	Elite + two_opt	1.00	0	0.1799
14	2,649.00	5.00	+2.7142%	Segment Crossover + Fast 2-Opt + weak_edge_surgery	1.00	0	0.2681
15	2,579.00	70.00	+0.0000%	Elite + two_opt + Selective LKH	1.00	0	29.8534

6. Final Discovered Tour & Solution Vector

Optimized Tour Length: 2,579.00 | **Optimality Gap:** +0.0000% | **Discovered at Generation:** 15

Permutation Sequence Preview:

```
0 → 279 → 2 → 278 → 277 → 247 → 248 → 255 → 254 → 251 → 208 → 207 → 252 → 253 → 256 → 257 → 258 → 259 → 260
→ 261 → 262 → 263 → 264 → 265 → 139 → 140 → 147 → 148 → 138 → 137 → 136 → 266 → 267 → 135 → 134 → 268 → 269
→ 133 → 132 → 131 → ... [240 additional cities] → 0
```

7. Execution Event Log & Diagnostics

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• Initialized a280 (280 cities). Initial best cost: 2,975.00
• Gen 1: Discovered new global-best 2,970.00 (-5.00)
• Gen 2: Discovered new global-best 2,958.00 (-12.00)
• Gen 3: Discovered new global-best 2,956.00 (-2.00)
• Gen 4: Discovered new global-best 2,945.00 (-11.00)
• Gen 5: Discovered new global-best 2,932.00 (-13.00)
• Gen 6: Discovered new global-best 2,914.00 (-18.00)
• Gen 7: Discovered new global-best 2,902.00 (-12.00)
• Gen 8: Discovered new global-best 2,879.00 (-23.00)
• Gen 9: Discovered new global-best 2,871.00 (-8.00)
• Gen 10: Discovered new global-best 2,687.00 (-184.00)
• Gen 11: Discovered new global-best 2,672.00 (-15.00)
• Gen 12: Discovered new global-best 2,668.00 (-4.00)
• Gen 13: Discovered new global-best 2,654.00 (-14.00)
• Gen 14: Discovered new global-best 2,649.00 (-5.00)
• Gen 15: Discovered new global-best 2,579.00 (-70.00)
• Gen 15: 0.0% optimality gap achieved. Experiment completed.
```

8. Artifact Manifest & Research Reproducibility

All raw experimental records have been saved to runs/a280_1791577811_9c8912/ including summary.json, generation_history.jsonl, best_route.csv, and report.pdf.